

Yagna Datta Gottipati

Mobile Robotics | SLAM | Python / ROS 2

CONTACT

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LinkedIn profile

TECHNICAL SKILLS

Robotics & Software

Python, Linux, ROS 2 Humble, Nav2, TF2, OpenCV, Gazebo, RViz2, Cartographer

Control & Sensing

LiDAR, RGB-D, feedback control, sensors and actuators

Design and Analysis

SolidWorks, CATIA V5, Ansys, FEA fundamentals

CERTIFICATIONS

German A2
telc certificate

SIX SIGMA PRINCIPLES
Kennesaw State University

CATIA V5
Beginner to advanced; automotive and industrial design Udemy

Python Programming for Beginners
Udemy

Python for Robotics
The Construct

PROFESSIONAL SUMMARY

M.Eng. student in Mechatronics and Cyber Physical Systems with practical Python and ROS 2 experience in mobile navigation, camera perception and LiDAR control. Currently working on LiDAR based SLAM Map updating for changing indoor environments using Gazebo and RViz2 in ROS 2 Humble. Project experience includes navigation architecture, SLAM, sensor integration and feedback control.

EDUCATION

M.Eng. Mechatronics and Cyber-Physical Systems

TH Deggendorf, Technology Campus Cham
03/2024 - Present

B.Tech. Mechatronics | 06/2019 - 05/2023

Hindustan University, Chennai, India

PROJECTS & RESEARCH

LiDAR Based SLAM Map Updating for Changing Indoor environments

Ongoing research | ROS 2, Nav2, 2D LiDAR, occupancy-grid mapping.

• Developing a map maintenance pipeline to distinguish persistent indoor layout changes from temporary obstacles using LiDAR evidence collected across separate robot visits.

• working on a map updating policy based on ray casting, scan to map alignment checks and separate thresholds for confirming and reversing occupied and free space changes.

• Planning an integration with ROS 2 and Nav2, maintaining a fixed localization reference while updating a separate, versioned navigation map.

GitHub repository

Multi Goal Logistics AMR

Hybrid predictive DWA & safety reflexes | ROS 2 Humble, Nav2, TF2, Python, Gazebo.

• Developed a ROS 2 based multi goal AMR navigation framework, integrating Nav2 path planning with a custom predictive DWA velocity supervisor for dynamic obstacle tracking and collision-aware motion selection.

• Implemented LiDAR based dynamic obstacle tracking and constant velocity trajectory prediction, evaluating 625 candidate robot motions over a 3 second prediction horizon at 10 Hz for collision aware velocity selection.

• Integrated multi goal navigation and safety supervision in Gazebo, successfully completing navigation targets across 31.94 m in 155 seconds. Used ROS 2 bag recordings and telemetry analysis for system performance checks.

GitHub repository

LANGUAGES

German - A2 (telc)

English - C1

Telugu - Native

Extracurricular activities

NSS Cell Secretary

Hindustan University
07/2022 - 05/2023

Chief Sergeant at Arms

Rotaract Club, Hindustan
06/2022 - 05/2023

PROJECTS & RESEARCH

ROS 2 Reactive Wall Following

ROS 2 Humble, Python, LiDAR Perception, PID control, Gazebo.

• Developed a map less autonomous boundary tracking node in ROS 2 Humble controlled by a deterministic Finite state Machine, processing raw 2D LiDAR scans for real-time corridor navigation.

• Designed an Asymmetric Angular Shielding sector filter that separates wall tracking and frontal obstacle detection sectors to reduce false obstacle triggers from the tracked wall during left wall tracking, and this prevented the tracked wall from triggering false forward collision-avoidance interrupts.

• Coupled a closed-loop PID controller with cmd_vel actuation to maintain a constant lateral wall standoff distance, achieving smooth trajectory convergence and zero collision cornering in Gazebo environments.

GitHub repository

Vision-Guided Line Tracking

ROS 2, Python, OpenCV, LiDAR, Gazebo

• Developed a vison-guided autonomous navigation node in ROS 2 Humble, utilizing cv_bridge and OpenCV for real time path centroid extraction.

• Designed a closed loop PID visual controller that minimizes lateral pixel tracking error via continuous cmd_vel modulation, incorporating autonomous search and recovery states to handle visual occlusion and sharp turns.

• integrated 2D LiDAR distance monitoring alongside camera perception preempting visual tracking with emergency deceleration and obstacle avoidance when hazards enter the forward safety zone.

GitHub repository

EXPERIENCE

International Student Assistant

TH Deggendorf, Technology Campus Cham | 04/2025 – 03/2026

• Supported in campus activities and day-to-day administrative tasks.

• Assisted international students with enquiries and campus-related quarries.

• Coordinated with the campus intercultural team on organizing and actively conducting interactive sessions and international events for international students.

Production Assistant Intern

Radiant Electronics and Appliances Pvt. Ltd. | 06/2022 - 09/2022

• Supported machine operation and day-to-day production activities.

• Monitored machine performance and assisted with operational checks.

• Assisted with inspections, basic maintenance and daily production team tasks.